

Reporting Module — Scheduled Reports from Saved Dashboards

Field	Value
Product line	Sand Platform capability, surfaced in SandGen (Sand Generation) first
Owner	Mohamed Moanes (Functional Lead / acting Product Manager)
Status	Draft
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Skill used	sandplatform-prd, with sandgen-prd consulted for the surface and the tier question
Candidate plane (Sand fit)	plane:platform, surfaced through app:sandgen. Must be confirmed with architecture
Sand-fit depth	Full. Decided by the PRD Architect — see Confidence & Flags
Source brief	SB-20260908-reporting-module-v1 (contract v2, Status: Ready for PRD, Content mode: PM-provided, confirmed by the Product Manager 2026-09-08)
Declared scope (from brief)	Platform; application scope declared "not applicable — platform-level". This PRD agrees on Platform and partly overrules the second half — the setup screen is application-resident. See Sand Fit
Document control	v1, 2026-09-08. First version. No predecessor
Companion	PRD-reporting-module-v1-ANNEX — field dictionary, screen states, failure states, architecture reasoning, check methods, dependencies, source record
Readers	Business / leadership · Product Owner · UI/UX Design · Engineering · Client
Approver	[NEEDS INPUT: who signs this off – Functional Lead, CTO, or both]

Who should read what

You do not need all of it. Read your row.

If you are	Read	Skip
Business or leadership	Executive Summary, The Driver, Goals, Non-Goals, Pricing, Rollout	Everything technical
Product Owner	All of it	The Annex
UI/UX Design	Users & Scenarios, Design Deliverables, plus the field dictionary, screen states and failure states in the Annex	The architecture sections
Engineering	Requirements, Sand Fit, Multi-Tenancy, Data & Integration, Acceptance, plus the Annex	Pricing
Client-facing	Executive Summary, Users & Scenarios, Non-Goals	The architecture sections

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Executive Summary

What is being built. A way to turn a dashboard somebody already has into a report that arrives by email on its own. The person picks a report, chooses which sites or devices it covers, picks a PDF or an Excel file, sets a day and a time, and lists the addresses it goes to. After that the platform builds the file and sends it, and nobody has to open Sand that morning.

Whose problem this is. Three teams at one client account — Asset Management, O&M, and what the client calls its Technical Office — rebuild and re-send the same reports by hand every time they are needed. Sand ships report generation today, but it is manual to make and manual to send, so nothing repeats on its own.

The driver, in one line. An account at **Pilot stage** is waiting for this, and three of its teams are doing the work by hand meanwhile. That is an External driver, re-tested here rather than inherited from the brief.

What is explicitly not in scope. Reports built on customer-defined formulas, anything on mobile, any delivery route other than email, billing documents, and any chart or figure a dashboard cannot already show. Reports are snapshots of dashboards; this module does not invent report content.

How success will be proven. Twenty-three acceptance checks, every one observable, most at the interface rung because the whole module sits behind sign-in. One needs a person to look at a PDF and agree it resembles the dashboard it came from. No baseline exists for anything today, so every numeric target in Goals is marked proposed rather than known.

The dates that matter. None. No delivery date, target quarter or committed date appears in this document — a deliberate decision by the Product Manager on 2026-09-08. What the document does state is **order**: nothing delivers until Sand has a working email route, and the private and shared report types cannot work until the Admin Module gives each person an account.

What is being asked of the reader. From architecture: an answer to **whose authority renders a scheduled report**, which blocks the build, plus confirmation that a scheduler and an email route exist to build on. From leadership: whether this sells in the entry package or the custom one, and whether a report may be emailed outside the client's organization. From design: the screens, working from Design Deliverables.

Assumptions

- **The Admin Module and the dashboard builders are in user acceptance testing, not released.** Taken from the brief and the Product Manager on 2026-09-08 without independent confirmation. `09_shipped-state.md` (2026-08-09) still records both as in progress, so that file is behind him here.
- **The waiting account is at Pilot stage.** Carried from the brief, which recorded the stage on 2026-08-09. Not re-verified, at the Product Manager's instruction.
- **The Formula Builder is postponed**, so no report in this release can depend on a customer-defined value.
- **Sand has a mail abstraction but no delivered email route.** Taken from PRD-notification-center-v5. Treated as a dependency to confirm, not a fact to build on.
- **"Technical Office" is the client's own name for a team**, not a Sand role. Kept in the Product Manager's words.

1. Problem & Motivation

Reporting in Sand Generation exists and works. What it does not do is repeat. In the Product Manager's words on 2026-09-08, a user *"can only create it manually and can only send it manually not scheduled."* A report somebody needs weekly is rebuilt and re-emailed weekly.

The cost lands on three teams inside one client account, all on the client's side: **Asset Management**, **O&M**, and the client's **Technical Office**. Two map onto roles Sand's own material names — Asset Manager and O&M Manager, per `01_company-and-product-overview.md`. The third does not, and the wording is kept as the Product Manager said it rather than mapped onto something else.

The account is identified by **pipeline stage: Pilot**, per Idea Brief SB-20260908-reporting-module. It is the account Sand releases through before other clients, which is what makes its open asks load-bearing rather than anecdotal. The Product Manager adds that every kind of user will eventually use this module; these three teams are where the pain is sharpest, not the whole audience.

Doing nothing: the three teams keep rebuilding the same files for as long as they use the platform, and the ask stays open while both things it depends on sit finished and in testing.

In plain words: people are doing by hand, every week, something the platform could do by itself.

2. The Driver

- **YES — External** — an account at **Pilot stage** is waiting for this module, and three of its teams produce their reports by hand today. Evidence is the Product Manager's own, dated 2026-09-08: *"they are doing everything manually"* and *"we want them to use our reporting module instead."*
- **NO — Structural — not claimed.** Nothing Sand published promises scheduled reports. 01_company-and-product-overview.md sells pre-defined reports in Basic and custom ones in Plus; 05_pricing-packaging-competitive.md glosses its Advanced tier as *"Custom Reports & Billing."* Both are about **what** a report contains, never **when** it arrives. Manual creation shipped and works.
- **NO — Generative — available and deliberately not claimed.** The class is real: once a saved artefact can be rendered to a file and delivered on a timer, billing documents and O&M summaries want the same machinery. But External justifies this whole release, and claiming generative would license a general delivery framework nobody asked for. If a second consumer appears, that is the moment.

Brief re-test. The brief claimed External; this document reaches the same conclusion independently, having checked the four things the classification could have failed on. The account is identified by **stage**, never by name. The evidence is the Product Manager speaking, not the brief's research — an article about industry-wide reporting costs is context, not a customer asking. The content mode is **PM-provided**, so the claim is not a generated hypothesis. And the brief correctly kept his view that the module *should have shipped sooner* as context rather than a reason: a pipeline position is not a driver. Nothing changed, but the checking still had to happen.

In plain words: a customer already using Sand is waiting, and three of its teams are doing this work by hand. That is enough.

3. Goals & Success Metrics

Metric	Baseline → Target
Recurring reports that arrive without a person	0 today — nothing repeats on its own, per the Product Manager → [NEEDS INPUT: how many] [PROPOSED — not measured]
Reports the three client teams assemble by hand each week	Unknown — nobody has collected them → a reduction, unquantified
Applications reusing the capability	0 today → 2 when Sand Distribution starts, with no second implementation [PROPOSED — not measured]
Scheduled runs that deliver on the first attempt	No baseline — the capability does not exist. A bar rather than a target, named in Quality Requirements as Delivery reliability

No number here was invented. The first baseline is known because the Product Manager stated it; every target is proposed and needs his confirmation. The second row is the honest center of this section: nobody has asked the three teams for the reports they build today, so this module is specified against a described need rather than a measured one. Their files are the best available description of what it must produce.

4. Non-Goals / Out of Scope

Not doing	Reason	Where it belongs instead
Reports built on customer-defined formulas	The Formula Builder is postponed, so the values would not exist	The Formula Builder release — Requirements 12 to 18 of PRD-customizable-dashboards-v2
Any mobile surface	The mobile app is in design, and the Product Manager has displaced it behind this module	Its own brief, which he has asked for next
New report content — charts, figures or KPIs the platform cannot already show	Every report is a dashboard snapshot; inventing content here builds a second authoring tool	The Diagram and Widget Builders in PRD-customizable-dashboards-v2

Not doing	Reason	Where it belongs instead
Report building outside a dashboard	Same reason. If a person wants what no dashboard shows, they build the dashboard first	The dashboard builders, as above
Any delivery route other than email — in-platform inbox, download links, file storage	Email is what the waiting teams already use; more routes now is unrequested configurability	A later release. Note the tension: download links answer the attachment-size risk in Risks
Billing documents and invoices	A financial artefact with its own rules, not a dashboard snapshot	The Billing module, over the billing domain architecture (candidate: ADR-033)
Output formats other than PDF and Excel	The Product Manager named two; a comma-separated export also produces nothing from a chart-only dashboard	A later release; Requirement 4 makes a format a registration, not a rewrite
Recipient-side permission checks	Recipients are email addresses, not platform users, so there is nobody to check. Deliberate, and the largest risk in this document	Not a later release — an answer to the blocking question in Sand Fit
Editing or annotating a generated file	Nobody asked. A report is a snapshot	A later release, if anyone asks

In plain words: this schedules and sends reports made from dashboards that already exist. It does not invent report content, does not go to phones, sends nothing anywhere except email, and does not do invoices.

5. Users & Scenarios

Which role list this uses

This document builds its permission table on the Admin Module's **four** roles — Owner, Admin, Engineer, Technician — and **not** on the eight commercial role names in `01_company-and-product-overview.md` (General Manager, Asset Manager, Business Manager, Finance Manager, O&M Manager, Regional Manager, Associate, Technician). That is a decision: the four are what the software will enforce, and this document specifies software. Only Technician appears in both, and **no mapping between them is proposed** — a guess about who somebody is becomes a guess about what they may do.

A later Admin Module release also lets a client **create its own roles with chosen privileges**, at which point neither list is authoritative and the client's configuration is. So **every requirement below is written against a privilege, never a role name**, and the table is a starting configuration rather than the specification.

The privileges this module introduces

Privilege	What it permits
See system reports	Open the reports Sand ships and use them as a source
Create a private report	Create a report visible only to its author
Share a report	Make a report visible to others in the same client account
Schedule and switch off	Set a repeating schedule on a report, and stop one
Switch off another person's report	Stop a shared report somebody else created
Send outside the organization	Add a recipient whose address is not in the client's own domain
Read run history	See when a report ran, who received it, and whether it worked

Who may do what — this is the only copy

Action	Owner	Admin	Engineer	Technician
See system reports	Yes	Yes	Yes	[NEEDS INPUT]
Create a private report	Yes	Yes	Yes	[NEEDS INPUT]

Action	Owner	Admin	Engineer	Technician
Share a report inside the account	Yes	Yes	Yes	No
Schedule and switch off their own report	Yes	Yes	Yes	No
Switch off a shared report somebody else created	Yes	Yes	No	No
Add a recipient outside the client's organization	[NEEDS INPUT — leadership]	[NEEDS INPUT — leadership]	[NEEDS INPUT — leadership]	No
Change what a system report contains	No — nobody in the client account	No	No	No
Read run history for a report they can see	Yes	Yes	Yes	[NEEDS INPUT]

Three cells are deliberately unfilled and one row is deliberately all-No. A Technician sees only assigned sites and manages nobody, so whether they get reports at all is a product question, not a technical one. The outside-recipient row is a leadership decision: a report leaving the client's domain reaches lenders, owners or auditors, and that changes what a shared report means. The system-report row is this document's proposal — a client scopes, schedules and delivers a Sand-shipped report but cannot change its content, or "system report" means nothing. **Architect-proposed — confirm.**

What people are trying to do

- As an **Asset Manager**, have last week's portfolio summary in my inbox on Monday without opening Sand.
- As an **O&M Manager**, send the same site report to the same three people weekly and stop rebuilding it.
- As somebody in the client's **Technical Office**, produce one report now, for one site, with no schedule.
- As an **Owner or Admin**, stop a shared report going to the wrong people without hunting for its author.
- As anybody who set one up, find out that last night's report did not send, and why.

The main path through it

Open the reporting setup page → pick a type: **system**, **my reports** or **shared** → pick a report from that type's list, each sourced from a dashboard → choose what it covers: sites, devices, or the whole portfolio, within what the source allows → choose **PDF** or **Excel**, and for PDF whether it carries diagrams and tables, diagrams only, or tables only → set a repeating day and time, or ask for it once → enter the addresses → switch it on. The platform then builds and sends it, recording every run.

When things go wrong

Eleven situations and the required behaviour for each are in the Annex, authored there and nowhere else. They are requirements, not notes. The four carrying most weight: the source dashboard deleted while a schedule is live; the author of a shared report losing access to a site it covers; a file too large to email; and a run failing at three in the morning with nobody watching.

[Design-relevant] As authority drops, controls are **hidden, not greyed out**, and the layout must not leave a hole where a hidden control would have been — the same rule the Admin Module and the dashboards use.

In plain words: pick a report, say what it covers, pick a file type, set a day and time, list who gets it, switch it on. Everything hard in this document is what happens when one of those five things changes afterwards.

6. Requirements

6.1 Functional Requirements

The Source column says where each comes from. **PM** traces to the Product Manager's own words in the Idea Brief. **Dashboards PRD** and **Admin PRD** are aligned with PRD-customizable-dashboards-v2 and PRD-admin-module-

multi-users-v4 rather than restated from them. **New** is this PRD Architect's, and every one of those is **Architect-proposed — confirm**. Rationale is in the Annex.

The report and what it covers

#	Requirement	Priority	Source	Proved by
Requirement 1	The reporting setup page offers three report types — system, my reports (private to their author) and shared — as tabs, each listing the reports available in it	MUST / PO	PM	Check 1
Requirement 2	A report sources from exactly one saved dashboard , default or customer-built, and names that dashboard on screen and in the produced file	MUST / PO	PM + New	Checks 1, 13
Requirement 3	Coverage is chosen as particular sites, particular devices , or the whole portfolio , only within what the source dashboard permits. An out-of-scope request is refused where the data is fetched , not where it is displayed	MUST / PO	PM + Admin PRD	Check 2
Requirement 4	Output is PDF or Excel , each a registered renderer rather than a branch in report code, so a third format later costs a registration	MUST / PO	PM + New	Checks 3, 4
Requirement 5	For PDF, the person chooses diagrams and tables, diagrams only , or tables only , and the file contains only what was chosen	MUST / PO	PM	Check 3
Requirement 6	For Excel, each panel becomes its own sheet , and a diagram panel contributes the values behind the chart rather than a picture — so a dashboard with no tables still produces a usable file	MUST / PO	New	Check 4
Requirement 7	Values resolve through the same calculation path the dashboard uses , for the report's own period. No second calculation path is added	MUST / PO	New	Check 16
Requirement 8	Portfolio coverage is offered only where the source dashboard's scope can be expressed at portfolio level . Otherwise the option is not shown, and a request made by identifier anyway is refused	MUST / PO	New	Check 15

Scheduling, running and delivery

#	Requirement	Priority	Source	Proved by
Requirement 9	A report carries a repeating schedule — a date and a time — which runs while the report is active and stops when it is switched off	MUST / PO	PM	Check 5
Requirement 10	Every schedule resolves in a declared time zone , stated on screen. A report set for 07:00 arrives at 07:00 somewhere specific	MUST / PO	New	Check 5
Requirement 11	A report can be generated once, on demand , producing exactly one file and one delivery	MUST / PO	PM	Check 7
Requirement 12	An active switch per report. Off stops delivery; on resumes it and does not backfill runs missed while it was off	MUST / PO	PM + New	Check 6
Requirement 13	Delivery is to a list of email addresses , which need not belong to platform users	MUST / PO	PM	Check 8
Requirement 14	Email leaves through the platform's mail	MUST / PO	New	Check 9

#	Requirement	Priority	Source	Proved by
	abstraction , never a direct connection to a mail provider			
Requirement 15	Every file is produced through the platform's storage abstraction and kept for a period stated in the product rather than left to infrastructure. [NEEDS INPUT: how long is a generated report kept, and does any regulation apply]	MUST / PO	New	Check 18
Requirement 16	A run history per report: when it ran, what it covered, which file it produced, who received it, and whether it succeeded. Every delivery is recorded before sending, because email cannot be recalled	MUST / PO	New	Check 17
Requirement 17	A failed run records the reason and tells the report's creator . Retries are bounded; after the last the report goes inactive rather than failing silently forever	MUST / PO	New	Check 19

Authority, permissions and separation

#	Requirement	Priority	Source	Proved by
Requirement 18	Every action is gated on a privilege , never a role name, so a client creating or renaming roles later breaks nothing	MUST / PO	Admin PRD + New	Check 10
Requirement 19	A dispatch resolves its data against one declared authority , and that authority is named in the produced file . A reader can always tell whose view of the portfolio they hold	MUST / PO	New	Check 11
Requirement 20	When the declared authority loses access to a covered site, the next run does not send . The report goes inactive and the reason is recorded	MUST / PO	New	Check 12
Requirement 21	Deleting a dashboard names every scheduled report depending on it first , and how many. Deleting anyway is a second deliberate action, after which those reports go inactive with the reason recorded — they never send an empty or broken file	MUST / PO	New, aligned with Dashboards PRD	Check 13
Requirement 22	Editing a source dashboard takes effect on the next run , and the file records the dashboard's last-changed time so a reader can see why two periods differ	SHOULD / P1	New	Check 13
Requirement 23	Report definitions, generated files and run history are scoped to one client account and unreadable from another by any route	MUST / PO	Admin PRD + New	Check 14
Requirement 24	Where a person lacks a privilege the control is hidden, not greyed out , and the layout leaves no hole where it would have been	MUST / PO	Admin PRD + Dashboards PRD	Check 10

Consistency with what already ships

#	Requirement	Priority	Source	Proved by
Requirement 25	Every quantity with kilo/mega/giga multiples is stored in its base unit and scaled only when	MUST / PO	Admin PRD	Check 20

#	Requirement	Priority	Source	Proved by
	displayed — including inside the PDF and Excel file. The label is derived from the value, never stored separately			
Requirement 26	The report lists support choosing columns, filtering, searching and paging, with a page-size selector, a "showing X of Y" counter, and count badges on the three tabs	SHOULD / P1	Admin PRD	Check 21
Requirement 27	The manual report creation that already shipped is unchanged by this release. Scheduled reports are added alongside it	MUST / P0	New	Check 22

In plain words: build a report from one dashboard, say which sites it covers, pick PDF or Excel, set a day and time in a real time zone, list who gets it. Then: record every send before it happens, stop sending when the source disappears or the person loses access, say in the file whose view it is, keep each customer's reports entirely separate, and leave the manual reporting that already works alone.

6.2 Quality Requirements

The full table is in the Annex. Five bars are specified and testable: **delivery reliability** (a run either delivers or is recorded as failed with a reason — silent non-delivery is a defect); an **attachment ceiling** that fails a run visibly before sending rather than letting a mail provider drop the file; **separation**, structural and inherited from the Admin Module, never a query filter; the **mail route**, through the abstraction only and verifiable in code; and **authority at dispatch**, resolved at the run and never cached from creation.

Six bars were never specified by anyone and nothing was invented to fill them: file retention, generation time, concurrent runs, uptime, accessibility, and expected volumes of sites, devices, users and active schedules. Each is [NEEDS INPUT] in the Annex and each appears in Open Questions. **Volume is the one that matters most** — a scheduler cannot be sized against an unknown number of reports all set for Monday at eight.

6.3 Detailed Field Specifications

The field dictionary is in the Annex, authored there and nowhere else, covering the report definition, the schedule, the recipient list and a run record. Every field is marked Required, Optional or System-managed, and quantities follow Requirement 25.

6.4 Design Deliverables

[Design-relevant] Mandatory — this module is mostly screen, and the brief is right that it is a lot of screen for one designer.

- **Screen inventory.** A new **reporting setup page** with three type tabs and the report list; a **report editor** for coverage, format, schedule and recipients; a **run history** view; and an **entry point from a dashboard** — "make a report from this". Plus the produced artefacts, which are design surfaces even though nobody clicks them: **the PDF layout** in its three content variants and **the Excel workbook** structure. **The delivery email is a fifth surface and needs a design, not a default.**
- **State matrix.** Per screen: empty, loading, forbidden, error, partial data. Authored in the Annex.
- **Interaction notes.** The hardest interaction is **coverage** — the picker must offer only what the chosen source dashboard permits, and it changes when the source changes. Second hardest is **making failure visible without alarming anyone**: seeing that last night's run failed, without the page becoming an incident dashboard. Left to design: the setup page layout, how the tabs present, how the PDF variants look. **Must not be invented:** the three report types, the three coverage kinds, the two formats, the three PDF content choices, the privilege table above, and the hidden-not-disabled rule.
- **Open design decisions.** Whether the schedule is set on the report or in a separate step; whether the dashboard entry point opens the setup page or a short form; what the recipient list looks like at twenty addresses; whether run history sits on each report or in one place.

In plain words: one new page, an editor, a history view, a way in from a dashboard — and the PDF, the spreadsheet and the email itself, which are designed things too.

7. Sand Fit — Placement & Mechanism Reuse

Sand-fit depth: Full. This PRD Architect's decision. This specifies behavior a developer implements against platform-dev: it reads device readings, resolves measures, renders files, writes to object storage and sends mail through an adapter.

Grounded in 08_sand-fit-architecture-digest.md, **generated 2026-08-02** — a dated snapshot of platform-dev, not a live connection. **Every claim here is a candidate that must be confirmed with architecture.** Nothing came from the brief; briefs carry no architecture by contract and this one honored that completely. **The full reasoning — package candidates, rejected placements, the mechanism mapping, the dependency argument and the decision table — is in the Annex.**

The conclusion: Platform, with an application-resident surface. The brief declared **Platform**, application scope "not applicable — platform-level". This document **agrees with Platform and partly overrules the second half.** Platform for three reasons, none from the brief: 01_company-and-product-overview.md lists **Reporting as a centralized backend engine shared across all frontends**; Sand Distribution's own reporting slot needs the identical capability; and PRD-customizable-dashboards-v2 already names *"the Report Creator"* as the candidate consumer of the platform's composition model, so building this inside Sand Generation would guarantee a second implementation. The overrule is narrow and real: **the setup page is application-resident**, in the Sand Generation shell, and the measure catalogue a Generator customer reports on is Generator vocabulary. "Not applicable" hides that, and confusing shared machinery with application vocabulary is how a second implementation gets built by accident.

Plane: `plane:platform`, surfaced through `app:sandgen`. One plane, not a seam — nothing reaches the assistant layer, since the Formula Builder is out of scope.

Reuse before invent — it shrinks the work. Five of the six fundamental mechanisms each carry a piece: the **adapter pattern** for mail and storage; the **polymorphic domain hierarchy**, because sites, devices and portfolio already *are* Scope; **JSON parameterization**, making a report definition configuration rather than schema; **automatic CRUD resource generation**, giving crud_resource for the definition and service_resource for dispatch; and **schema-driven dynamic views**, so rendering reads the same document-canvas-element chain the screen does and the file agrees with the dashboard by construction. The Annex maps each. **What this document adds is four things:** the report definition and its privileges, two renderers, a timer, and delivery. Everything else is registration.

In plain words: most of this already exists in Sand's shared code. The new parts are the report itself, the PDF and Excel writers, the timer, and the sending.

Extension without modification — yes, with one honest gap: the timer. The digest names EventBus for announcing that something happened, but a **time-triggered** scheduler appears nowhere — not in the six mechanisms, not in the verified class locations, and not in INTAKE_GUIDE.md's language table, which has a row for *"notify when X happens"* and none for *"every Monday at seven."* The core of this module is a timer, and the digest does not confirm one exists.

Dependency direction — one tension, named rather than resolved. Dispatch feels infrastructural, pulling it toward Foundations; but a report **reads measures**, which live in Data Processing (dataflow), so one low-layer component would depend upward, which is illegal. The likely split is a generic scheduling primitive low down and the definition above dataflow — architecture's call. Relocating the whole thing downward to make imports legal would be the wrong move.

Open placement questions for architecture

Name	Question	Blocks the build?
Whose eyes render the report	A dashboard resolves what you may see when you open it . An emailed file is never opened by anyone with permissions — it goes to an address. So whose authority renders it: its creator's, at the run? A named service authority? And what happens when that authority's access changes? Requirements 19 and 20 are this document's proposal; the question is whether the platform can express it	Yes — blocks everything about shared and scheduled reports
Whether a time-triggered scheduler exists	The digest names none. Does platform-dev have one, is one arriving with another module, or is building one this module's first	Yes — it is the core mechanism

Name	Question	Blocks the build?
	task?	
Whether an email route exists to build on	PRD-notification-center-v5 records a mail <i>abstraction</i> but no delivered email route , in the pipeline with no date. This module has exactly one delivery channel	Yes — nothing delivers without it
Where a report definition lives	Configuration under JSON parameterization, or new tables? This inherits the unresolved " <i>where a saved view lives</i> " question from PRD-customizable-dashboards-v2, and it should be answered once for both	Yes — it decides whether there is a migration at all
Where the file is rendered, and by what	Nothing in the digest names a PDF or spreadsheet renderer. Does one exist, and does rendering run in the request path?	Yes — for the renderers
What `packages/analytics` is	In the digest's verified file locations, in none of its four package layers, and a likely home for part of this. PRD-customizable-dashboards-v2 raised it and it is still open	No — but part of the placement argument

8. Multi-Tenancy, Apps & Composition

A platform capability with an application-specific surface — the configuration that most often needs an architect. The building, scheduling and sending are composed by every Sand application; the setup page and the Generator measure catalogue are Sand Generation's.

Separation is structural and inherited, not invented here. This module adds no separation mechanism. It relies on the Admin Module's model: a client identifier on every top-level record, a per-client sign-in space, and site access cascading Company → Sub-Company → Site → Device. Definitions, files and run records are client-scoped records like any other.

Where the risk actually is — three failures, in descending order of damage.

1. **A file reaching an address it should never have reached.** The new route this release opens, and worse than anything the dashboards added, for one reason: **email cannot be recalled**. A dashboard shown to the wrong person can be un-shared; a PDF of a client's portfolio sent to a mistyped address is gone. Requirements 13, 16 and 19 exist for this, and it is why the outside-recipient privilege is not decoration. 2. **A scheduled report continuing to send data its authority may no longer see.** The realistic one, because nobody has to do anything wrong: access is narrowed for good reasons and a months-old schedule keeps running. Requirement 20 stops the run rather than trimming the file, deliberately — a report that silently shrinks is one nobody can trust. 3. **Portfolio coverage quietly including a site the authority is not entitled to.** The subtle one: nothing looks wrong, the file renders, and the number is computed over more sites than it should be. PRD-customizable-dashboards-v2 put it best and this document does not re-argue it — **a number is as disclosive as a chart**. Requirement 3 enforces it where the data is fetched; Check 2 tests it there.

Composability. Because this registers into existing composition machinery rather than sitting beside it, Sand Distribution's reporting consumes the same capability with its own measure catalogue when development starts — expected around September 2026 per 09_shipped-state.md (2026-08-09). That is the reuse metric in Goals. **Cross-application visibility: none**; Sand Distribution has not started.

In plain words: each customer's reports are walled off by the same structure that walls off their sites. The new danger is that a file leaves Sand by email and cannot be taken back.

9. Data & Integration

Routing was derived from the Product Manager's **verbatim** words as quoted in the brief, matched against the language table in INTAKE_GUIDE.md. **The quotations are reproduced in full in the Annex**; they are the only record of the original phrasing.

The Product Manager said	Where it routes
<i>"reads all kind of renewable energy Sites Readings ... energy and power readings"</i>	The dataflow package — DataAcquisitionService, the ReadingFrame family, time-series storage
<i>"& KPIs"</i>	The measure pipeline (candidate: ADR-035). Note: the digest records two files numbered ADR-031 , one on measure typing and aggregation, so decisions here are cited by filename and never by number alone
<i>"in forms of Widgets and Diagrams and tables"</i>	Schema-driven dynamic views; the subject-display read-side element (candidate: ADR-029), which the digest places in the frontend
<i>"exported from the system default dashboards and custom dashboards"</i>	The document-canvas-element chain PRD-customizable-dashboards-v2 specifies. This module consumes it and adds nothing to it
<i>"select the applied sites, devices or even on the portfolio"</i>	The polymorphic Scope hierarchy, with hierarchical scope authorization (candidate: ADR-024)
<i>"PDF or Excel Sheet"</i>	Object storage through the storage abstraction, MinIO-backed. Two registered renderers
<i>"put all the mail/s"</i>	The mail abstraction through the adapter pattern (candidate: ADR-006). Never a direct provider connection
<i>"sent automatically in the scheduled date and timing"</i>	No row exists for this. The table covers <i>"notify when X happens"</i> — an event — and nothing covers a timer. This is the gap named in Sand Fit

In plain words: readings come from the part of Sand that already handles device data; the layout comes from the dashboards; the file goes to Sand's file store; the email goes out through Sand's mail layer. The only thing with no home yet is the clock.

Everything is behind sign-in, which sets the minimum verification rung for anything touching it (candidate: ADR-023, the cookie-based sign-in decision). Interface-rung checks are the floor, not unit tests. **Two external systems**, both adapter-mediated: the mail provider and object storage. No third-party-facing interface is proposed — Sand's public interfaces are in progress per 09_shipped-state.md and nothing here needs one. **Migrations: additive only**, and whether there is one at all depends on where a report definition lives. No destructive operation is proposed.

10. Acceptance Criteria & Verification

Every check below is Architect-proposed — confirm — the whole ladder is this document's proposal, not the Product Manager's. Nothing came from the brief; briefs carry no verification rungs by contract. **How each check is performed is in the Annex**; what each proves is here.

#	Acceptance check	Proves	Verified at
Check 1	A report is created against a saved dashboard and persists with its type, source, coverage, format, schedule and recipients	Requirements 1, 2	API
Check 2	Coverage outside the source dashboard's permitted scope is refused where the data is fetched , including when asked for by identifier	Requirement 3	API
Check 3	A PDF is produced in each of the three content choices and contains only what was chosen	Requirements 4, 5	API
Check 4	An Excel file carries one sheet per panel, and a diagram-only dashboard still produces series values rather than an empty workbook	Requirements 4, 6	API
Check 5	A repeating schedule fires once per period in its declared time zone — a controlled clock advance produces exactly one dispatch, not two and not none	Requirements 9, 10	CLI, then API
Check 6	Switching off stops dispatch; switching on resumes without	Requirement	API

#	Acceptance check	Proves	Verified at
	backfilling missed runs	12	
Check 7	A one-time request produces exactly one file and one dispatch	Requirement 11	API
Check 8	Every recipient receives the file, including an address belonging to no platform user	Requirement 13	API
Check 9	Mail leaves through the mail abstraction, with no direct provider call anywhere in the path	Requirement 14	CLI
Check 10	A person without a privilege is refused at the interface, and the control is absent from the page rather than disabled	Requirements 18, 24	API, then Playwright
Check 11	A shared report renders against its declared authority, and the file names that authority	Requirement 19	API
Check 12	When the authority loses access to a covered site, the next run does not send, the report goes inactive, and the reason is recorded	Requirement 20	API
Check 13	Deleting a source dashboard names every dependent report first; deleting anyway makes them inactive with a recorded reason and sends no file	Requirements 2, 21, 22	API
Check 14	A report, its files and its run history are unreadable from another client account by every route tried	Requirement 23	API
Check 15	Portfolio coverage is not offered over a single-site source dashboard, and is refused if requested anyway	Requirement 8	API
Check 16	Values in a produced file match the source dashboard for the same coverage and period — one calculation path, not two	Requirement 7	API
Check 17	Every dispatch is recorded before sending, with its recipients and outcome	Requirement 16	API
Check 18	A generated file is stored through the storage abstraction and is retrievable afterwards	Requirement 15	API
Check 19	A failed run records its reason, retries a bounded number of times, then goes inactive and informs its creator	Requirement 17	API
Check 20	Quantities in the PDF and Excel are scaled from the base unit, with the label derived from the value	Requirement 25	API
Check 21	The report lists support columns, filtering, search, paging, page size, the counter and the tab badges	Requirement 26	Playwright
Check 22	Manual report creation still works exactly as before, alongside the new path	Requirement 27	Playwright
Check 23	A PDF is recognizably the dashboard it came from — layout, ordering, legibility, nothing cut off	Requirements 4, 5	Operator-confirm

Why the rungs sit where they do. Everything is behind the proxy and the identity system, so the interface rung is the floor, not the ceiling — a unit test cannot see a cookie problem. Two checks drop to the command line for something a live request cannot give: Check 5 needs control of the clock, Check 9 needs to see that no provider call exists rather than that one succeeded. Three climb to the browser because they concern what a page renders. **One check is operator-confirm and only one:** whether a PDF looks like its dashboard is an aesthetic judgement genuinely beyond autonomous reach. It **gates the release close, not the dispatch** of work, and stands in for nothing a machine could check.

Traceability, stated outright. Every MUST has at least one check. **Two requirements are not fully verifiable:** Requirement 22's last-changed timestamp is only partly covered by Check 13, and Requirement 15's retention period cannot be checked until somebody states it — Check 18 tests the mechanism, not the duration.

In plain words: almost all of this can be proved by a script hitting the real system. Two things need a controlled clock. One needs a person to look at a PDF and say whether it resembles the screen it came from.

11. Pricing / Packaging Impact

*The row used is `05\_pricing-packaging-competitive.md`'s SandGen table, whose upper tier is named Advanced and glossed "Custom Reports & Billing."*** That gloss is the closest thing in Sand's pricing material to this module.

But this module straddles the line, and that is a finding rather than a difficulty. 01_company-and-product-overview.md puts **pre-defined** reports in **Basic** and **custom** reports in **Plus**. This module does both: it schedules reports sourced from **system default dashboards** — pre-defined, Basic-shaped — and from **customer-built dashboards** — custom, Plus-shaped. And **scheduling itself appears in neither tier list**. Whether automatic delivery is an entry-tier convenience or the thing the upper tier is for **is not documented anywhere and cannot be looked up**. It is a commercial decision.

Tier-name caution. 01 packages SandGen as Basic / Plus / Gen Add-Ons; 05 prices it as Basic / Advanced, and **nothing states that Plus and Advanced are the same tier** (KNOWN_SOURCE_AMBIGUITIES.md A1). The equivalence is treated as **assumed and undocumented**; the names are not used as synonyms.

No bracket shift. Nothing changes installed capacity or zone, so nobody crosses the sub-1 MWp, 1-to-100 MWp or 100-MWp-plus boundaries, and the hybrid and micro-grid ten-percent add-on is unaffected. Both 01 and 05 are **September-2025** snapshots; confirm current rates before quoting externally.

One thing Commercial should know. The waiting account is at Pilot stage and **whether its package includes the tier this lands in is recorded nowhere**. If it does not, the External driver and the commercial answer point in different directions, and somebody should know before the conversation rather than during it.

12. Risks, Open Questions & Dependencies

The one question filed as blocking

Whose eyes render the report. Everything else here has a proposal attached. This one's proposal — Requirements 19 and 20 — may not be expressible in the platform as it stands, and the whole shared-and-scheduled half of the module rests on it. A dashboard decides what you may see at the moment you open it. **An emailed file has no moment of opening**, so the authority must be decided at the run and carried in the file. Until architecture confirms that, the private-report path can be built and the shared path cannot.

The risks worth acting on

Risk	What to do about it
There is no delivered email route, and this module has exactly one delivery channel. PRD-notification-center-v5 records email as in the pipeline with no date	Treat the email route as a dependency to confirm before build , not an assumption. Sequence the renderers first so the core can be proved without it — see Rollout
A file leaves Sand and cannot be recalled. One typo in a recipient list sends a portfolio to a stranger	Requirement 16's record-before-send, plus a leadership answer on outside-organization recipients. Both cheap now, impossible later
The single designer. This is mostly screen, and it has already displaced the mobile app in the design queue by the Product Manager's own decision	Design Deliverables names what must not be invented, which is where a designer's time goes. 09_shipped-state.md records roughly six engineers and one designer
A release-and-verification queue, not a build queue. 09_shipped-state.md (2026-08-09) records three things built but not released; this module's two dependencies	Do not read "the dependencies are finished" as "the dependencies are available." That distinction is the whole risk

Risk	What to do about it
are in testing, drawing on the same engineers	
Nobody knows the volume. No figure exists for sites, devices, users or active schedules	A scheduler sized against unknown load is a guess. Ask before building the timer
Attachment size. A portfolio Excel file may exceed what a mail provider accepts, and download links are explicitly out of scope	The attachment-ceiling bar fails the run visibly rather than losing the file silently. If it bites often, the download-link non-goal is the first thing to revisit

The full risk table and the dependency table are in the Annex.

What this waits on

Three things, and only the first two were in the brief: the **Admin Module** (private and shared reports mean nothing until each person has an account), the **dashboard builders** (they produce the sources), and a **working email route** (the delivery channel). The first two are in user acceptance testing per the Product Manager on 2026-09-08 — newer than 09_shipped-state.md, which still records both as in progress. The third is in the pipeline with no date. **The full dependency table is in the Annex.**

Roadmap position, re-checked against 09_shipped-state.md (2026-08-09) rather than taken from the brief. **G1.3 Reporting is Shipped** — *"users can generate reports in SandGen"* — and the Product Manager confirms what shipped is manual creation and manual sending, so this module automates what that slot left manual rather than replacing it. **G1.7 Reporting + Mobile is In progress** with its reporting half **not separately confirmed**, so whether this closes that half is unknown rather than assumed. **D0.5 O&M, Billing & Reporting** for Sand Distribution is **Slipped**, development not started.

Rejected alternatives are in the Annex, including the two the upstream Project generated for the Product Manager to react to and the smaller first release he declined.

Destructive-operation note: none proposed. Everything is additive. The one place data disappears is a file passing its retention period — a derived artefact, not a record — and that still needs the period answered first.

In plain words: the biggest risk is that the module's only delivery route does not exist yet. The most urgent question is whose permissions a scheduled report is built with.

13. Rollout & Phasing

Additive throughout. Manual report creation stays exactly as it is and nothing is removed — Requirement 27. **No dates appear here, by the Product Manager's decision on 2026-09-08. These are stages in order, not a schedule.**

Stage	What lands	Shippable on its own?
Stage 0 — the prerequisite	A working email route exists and is verified. Not this module's work , but nothing in it delivers without it	A gate, not a stage
Stage 1 — the renderers, proved without delivery	One-time generation from a system default dashboard, single-site coverage, PDF only. The file is produced, stored and retrieved. No scheduling and no email	Yes, and this is the important one. It proves the hardest new machinery — a dashboard becoming a faithful file — without waiting on the email route or the Admin Module. Checks 3, 16, 18, 20 and 23 all run here
Stage 2 — delivery	Recipient lists and one-time delivery by email through the mail abstraction, with run history recording every dispatch before it happens	Yes, once Stage 0 is in
Stage 3 — the timer	Repeating schedules in a declared time zone, the active switch, bounded retries, failure notification	Yes

Stage	What lands	Shippable on its own?
Stage 4 — private and shared reports	The three type tabs, the privilege table, the declared authority in the file, and the access-loss and dashboard-deletion behaviors	Needs the Admin Module released, and needs the blocking question answered
Stage 5 — the rest of the shape	Excel output, the three PDF content choices, device and portfolio coverage	Yes

Why Stage 1 is first and not the schedule. The timer is the most visible thing missing and not the riskiest thing to build. Turning a dashboard into a file a person recognizes is the part that can fail quietly and expensively, and the only part depending on nothing outside this module. Proving it first stops the email and Admin Module dependencies from blocking progress.

Rollout control. Per-client, through the entitlement machinery platform already owns, so the module can be switched on for the waiting account before it is general. **Architect-proposed — confirm;** nobody specified a rollout mechanism. **Sizing: not estimated** — the two things that would move any estimate, whether a scheduler and a renderer exist, are open questions.

14. Decomposition Hint

A starting point for the Planner, not a breakdown.

- **Candidate epic title:** Scheduled reports from saved dashboards
- **Suggested priority: P1.** The driver is External with a real account at Pilot stage, which would ordinarily argue P0 — but no date is committed and nothing delivers until an email route exists that is outside this document's control. P1 with an unblocking task first is honest; P0 on a gated capability just makes the gate invisible.
- **Plane tag:** plane:platform, surfaced through app:sandgen
- **Skill hints:** skill:pkg-dataflow, skill:pkg-platform, skill:mech-adapter, skill:mech-crud-resource, skill:mech-schema-driven-views
- **Acceptance class:** accept:autonomous, with a single accept:operator-confirm gate on PDF fidelity (Check 23) at release close
- **What the first task should be: not a schema.** Answer *whose eyes render the report*, and confirm whether a scheduler and an email route exist. Three questions, all answerable in a conversation, each changing what gets built.

15. Confidence & Flags

The full record is in the Annex — what is solid versus assumed, source ages, the ambiguities and how each was handled, brief provenance, every overrule, the auto-draft disclosure and the marker record. Restated here is only what somebody has to act on.

Must be confirmed with architecture, by name. Five block the build.

- **Whose eyes render the report** — the declared authority for a scheduled dispatch. **Blocks shared and scheduled reports entirely.**
- **Whether a time-triggered scheduler exists** in platform-dev today. **Blocks the core mechanism.**
- **Whether an email route exists to build on** — a mail abstraction is named; a delivered route is not. **Blocks all delivery.**
- **Where a report definition lives** — configuration or new tables. **Blocks the schema decision**, and it is the same question PRD-customizable-dashboards-v2 left open about saved views. Answer it once.
- **Where and by what a file is rendered.** **Blocks the renderers.**
- **What `packages/analytics` is** and which layer it belongs to — part of the placement argument, not a blocker.
- **The dependency-direction tension** between dispatch wanting to sit low and measure-reading sitting high. Do not resolve it by relocating downward.
- Every package, mechanism and decision named in Sand Fit and Data & Integration is a **candidate**, not settled fact.

Must be confirmed with the Product Manager, leadership or Commercial.

- **Whether a report may go to an address outside the client's organization** — the privilege row left unfilled in Users & Scenarios.
- **Which package this sells in**, given that it straddles pre-defined and custom content and that scheduling is in neither tier list.
- **Whether the waiting account's package includes that tier.**
- **How long a generated file is kept**, and whether any regulation applies.
- **Whether a Technician uses this module at all** — three cells in the permission table.
- **What the three client teams build by hand today.** Nobody has collected their files, and they are the best available description of what this must produce.
- **Whether this closes the reporting half of the G1.7 roadmap slot** or sits beside it.

Snapshot caveat. Architecture grounding is 08_sand-fit-architecture-digest.md, **generated 2026-08-02** — a dated snapshot of platform-dev, not a live connection, recording a **live duplicate ADR-031 collision**. One half of it, measure typing and aggregation, is load-bearing in Data & Integration, so every decision reference here must be confirmed **by filename** against the live repository, never by number. Commercial grounding is 01_company-and-product-overview.md and 05_pricing-packaging-competitive.md, both **September-2025** snapshots. 09_shipped-state.md is **last updated 2026-08-09**, wins over the older roadmap wherever it speaks, and is **behind the Product Manager** on the Admin Module and the builders. **The live repository and the current price sheet are the final authority, not this document.**

Known source ambiguities touched, none resolved silently. Two: the **two tier vocabularies** (A1), handled in Pricing rather than picked between; and Sand's **two live role lists** (A7), where the permission table uses the Admin Module's four, no mapping was invented, and every requirement is written against a privilege. The Annex gives the handling for each.

Open Questions

Architecture

- Whose authority renders a scheduled report, and what happens when it changes? — **blocks shared and scheduled reports**
- Does a time-triggered scheduler exist in platform-dev today? — **blocks the core mechanism**
- Is there a delivered email route, or only the mail abstraction? — **blocks all delivery**
- Does a report definition live as configuration or in new tables? — **blocks the schema decision**
- Where is a PDF or spreadsheet rendered, and does it run in the request path? — **blocks the renderers**
- Does a report read values a dashboard already resolved, or re-resolve them for its own period? Requirement 7 proposes re-resolving through the same path; confirm it.
- What is packages/analytics, and which package layer does it belong to?
- How is the dispatch-low, measures-high dependency tension split?
- Which of the two files numbered ADR-031 governs the measure work here?

Product and design

- Is the schedule a property of one report, or a reusable object several reports share? This document proposes the first, matching the Product Manager's flow.
- Which repeat intervals are offered — daily, weekly, monthly, or a stated set?
- Does run history belong on each report or in one place for all of them?
- Does the entry point from a dashboard open the setup page or a short form?
- What does the recipient list look like at twenty addresses?

User or customer

- What reports do the three client teams actually build by hand today?
- Do recipients include people outside the client's organization — lenders, owners, auditors?
- Has the account's pipeline stage changed since it was recorded as Pilot on 2026-08-09?

Leadership and commercial

- Is this module Basic or Plus, given that it straddles pre-defined and custom content and that scheduling is in neither list?
- Does the waiting account's package include the tier it lands in?
- May a report be emailed outside the client's organization?

- Does this close the reporting half of G1.7, or sit beside it?
- Who approves this document?

Data nobody has

- How many sites, devices, users and active scheduled reports should this be built for?
- How long is a generated file kept, and does any regulation apply?
- What is the attachment size ceiling before a mail provider refuses?
- Should a diagram-only report in Excel carry the values behind the charts? Requirement 6 proposes yes.
- Is "Technical Office" a role the platform should know about, or a client team that maps onto roles Sand already has?